On Thought, Language and Language of Thought: Critical Assessment of the Relation of Language and Thought from Evolutionary Perspective

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Abstract

The relation of language and thought is a broad and contentious topic, where the Language of Thought Hypothesis defining human thought as a computing operation on conceptual primitives, well known in the literature as an attempt to offer a formal theory of human reasoning, has been central to debates. It is embraced by the generative/biolinguisic perspective where a computing mechanism of higher level of abstraction, Universal Grammar, is incorporated in the formalism, as a crucial stage in the evolution of the language faculty as a regulator of silent reflection and a cornerstone of human speciation. Referring to expert findings in neurobiology, cognition, evolution etc. the article challenges the argument for the origin of language as computational mechanism , outlined by the LoT hypothesis. Inconsistency with the nature and functions of the brain as a bio-cognitive entity in evolutionary context, its misrepresentation of human thought as a digital operation and its inconsistency with evolutionary mechanisms are pinpointed and contrasted with alternative proposals. The article demonstrates that the relationship of thought and language is one of mutual influence which has evolved throughout human history.

Keywords: Language of Thought(LoT) hypothesis, Merge, evolution of language

1. Introduction : the relation of language and thought, a contentious topic

The relationship of language and thought is a broad topic and a rich literature covers diversity of perspective and viewpoints. That language influences thought , in the sense that the information we receive by communicating through using language influences our decision-making on a daily basis is not in dispute and is not discussed here. The disputes among philosophers, cognitive scientists, linguistics and others is the nature and the extent of this relationship, where some argue that the essence of human reasoning is in its reliance on language, strongly suggesting the absence of complex thought in languageless minds, while others see language as only one contributing factor where access to language injects clarity in one's thinking process, primarily attributed to the use of symbols as linguistic signs. The issue is further complicated by differences in identifying the defining property of language as it is defined by some as structure/syntax and as meaning/lexicon by others. In this sense some argue that grammatical categories and their structural arrangements in language determine the structure of human thought and the mind's perceptions of reality, e.g. the perception of time is influenced by the grammar of tense. Others suggest that the semantics of a language shapes one's thoughts and perceptions in interaction with reality, e.g. the calendar influences our perception of the weather and spatial vocabulary influences perception of space. In addition, the direction of influence, e.g. whether thought influences language or vice versa is also in dispute. For a condensed but illuminating summary of views and perspectives see Bloom and Keil (2001). And despite the diversity perspectives are clustered around two main themes.

1.1. Universal aspects of thought and language universals

Various scholars recognize the existence of universal properties of mind which precede and underly language attainment, although there are significant differences in defining these properties. Some point at causal reasoning and social cognition as part of general intelligence as universals of thought, which become enhanced by language attainment. Others offer a formal theory of human thought in terms of a cognitive algorithm, or Language of Thought, closely resembling artificial systems, a uniquely human properties of human minds, a defining feature of human cognitive exceptionality and the hallmark of human speciation.

1. 2. Language diversity and diversity of thought: the Sapir-Worf hypothesis

The Sapir-Worf Hypothesis stipulates that cultural differences reflected in the diversity of language systems can explain some cognitive and perceptual differences, e.g. in many languages the vocabulary of professions is usually marked by male gender markers in reflection of sexist attitudes in the workplace and the broad society, etc. Others argue that cultural influences are only marginal and perception is usually a universal property, e.g. colour perception is universal and largely unaffected by cultural idiosyncrasies.

The present article will focus predominantly on the LoT hypothesis and its claims for the nature of human reasoning and its role in the evolution of language. It summarizes findings which challenge the hypothesis from a number of perspectives and suggest that language and general cognition are processed independently by different mechanisms at different brain locations, thus challenging the LoT hypothesis. The article demonstrates that both universal properties of language and idiosyncrasies of languages influence human reasoning, in agreement with Bloom and Keil (2001) although the main focus of the article will be on the universal aspects of this relationship. The Sapir-Worf hypothesis will be mentioned only in passing for illuminating the role of individual language systems in shaping divergent representations of semiosis and perceptions of reality. In addition the article argues that the relation between language and thought has evolved during human evolution, e.g. at the onset of humanity language has assumed a marginal role in coding human semiosis, while assuming a leading role with the advent of civilization and writing, where language through education and literacy imposes the structural properties of written text on our stream of thought.

2. The Language of Thought hypothesis and its challenges: human reasoning and artificial systems

The argument for human thought as a computing operation on conceptual primitives, labeled as Language of Thought (LoT) is well known in the literature as an attempt to offer a formal theory of human reasoning, where human reasoning is defined as language, internal to the brain, i.e. human thought is envisioned in terms of semantics and syntax. In this context intentional mental states with truth conditions are sentence-like (Fodor 1975 and elsewhere). The so hypothesized LoT is regarded primarily as a mechanism for computing hierarchical organization of human thought with a typical predicate-argument structure. One would describe LoT or mentalese as a mental language where innate conceptual primitives are combined under abstract rules and form complex propositional hierarchically organized thought. The hypothetical cognitive primitives, e.g. actions, individual entities, properties, relations etc. are represented in the mind as generalizations , i.e. independent of perceptual experiences. The structural organization, i.e. syntax, of a proposition is independent of the conceptual content which fills the syntactic positions. In this sense the LoT is a mental algorithm, which computes sentence-like thoughts instantiated, or "written" in neuronal networks designed for brain-internal communication. It is understood as unique to the human species and universal, i.e. all minds have exactly the same concept inventories and the same rules. The abstract rules or the " laws of thought" are said to be similar but not identical to the rules of logic, e.g. logical operators (negation, quantifiers, etc.) form part of the structure of thought. In short, the LoT hypothesis defines meaning and human reasoning as abstract, that is, independent of perceptual experience, innately predetermined, universal, and digital. Importantly, LoT is a spatially isolated and functionally distinct bioprogram for computation of meaning, encapsulated within the brain anatomy and insulated from the rest of its cognitive functions, e.g. from lower level cognitive processes like spatial orientation, perception, motor control, etc.

To remind, the LoT hypothesis is a purely formal conceptualization of the human mind. Attempts have been made by the evolutionary psychologists Tooby and Cosmides(1994 and elsewhere) to provide empirical evidence for compartmentalization of mind and for the LoT, as formally conceived. Nevertheless, in all life forms cognition must reflect experience because cognitive capacities exist and evolve only if relevant to survival. Defining human cognition as independent of perceptual reality and experience with it and in terms of modern artificial systems makes this argument implausible for most scholars. The unprecedented cognitive entity is said to be a computation mechanism, capable of deriving all concepts ever conceivable by human minds from a small number of abstract conceptual atoms, organized in structured combinations under abstract rules (Fodor 1975, Fodor, Pylyshin 2015). In short, the LoT hypothesis defines meaning as abstract, that is, independent of perceptual experience, innately predetermined, universal, and digital, produced by processing mechanisms highly unusual for a bio-cognitive entity.

From evolutionary perspective Reboul (2015) justifies the claim for the human cognitive exceptionality with the stipulation of the exceptionality of human concepts as a departure from the conceptual systems of great apes. The LoT as a novel type of cognitive processing is said to initiate the sudden transformation of the ape brain into human hypothesizing a " Great Leap Forward" said to initiate the birth of new cognitively distinct, knowledge seeking Sapient species. Moreover, despite its cognitive sophistication, the LoT is assumed to have appeared in the human mind in its complete form at once, without precursors, an evolutionary Big Bang, diverging from fundamental evolutionary principles of continuity.

A reinvention of LoT hypothesis by acknowledging the role of perception in cognition as the two systems interface and attribute LoT-like properties of perception is proposed by Quilty-Dunn, Porot, Mandelbaum (2022). Moreover, it is argued that elements of LoT are found in pre-linguistic infants and non-human primates, departing form the original stipulation of LoT as uniquely human aspect of cognition, suggesting cognitive continuity. In addition, it is acknowledged that human cognition is more than symbol manipulation and incorporates a role for learning and experience.

" We believe in other vehicles of thought, associations, icons and much more.(ibid. in the Conclusion)

The hierarchical organization of human thought, or LoT is said to be the evolutionary foundation for the Language Faculty (LF) purported to be the result of the accidental association with another cognitive innovation, Universal Grammar (UG) a grammar-processing algorithm. UG is envisioned to have appeared in the brain as a resolution of a brain-internal conflict between the increased anatomical volume of the brain and the spatial constraints of the skull. As a result, grammatical concepts and computational principles were imposed on LoT, producing a highly complex computing entity for processing linguistic meanings and their configurations. Thus, in generative/biolinguistic context the formation of the LF is envisioned in terms of adding computational sophistication to human reasoning in terms of syntactic categories and highly abstract rules of their association. This justifies the apparent complete and total overlap between the structural organization of propositional thought, i.e. LoT, and the structure of language or UG.

The hypothesis of LoT is congruent with the generative perspective which views human language processing as computational, i.e. as automatic assembly of digital symbols into hierarchical structures, which furnishes the innatist argument that the human brain is equipped with innate computing mechanism containing the language algorithm, first articulated by Chomsky in his earliest works in the 60s and maintained to the present. In this sense the language faculty is a property of the individual mind, a cognitive hardware installed in the human brain with the function of organizing the rational thoughts of the human individual, or individual's silent reflection, where the mind is focussed on its internal life, distanced from experiences with the world and interpersonal interactions. Within the generative/biolinguistic paradigm LoT is positioned as Logical form in direct association with UG, the grammar algorithm and its relation to communication is indirect by attachment of computing mechanism of phonological form and its material articulation predominantly by speech.

Mirroring Chomsky's contention with behaviourism in linguistics in the 60s Fodor(1998) articulates his contention with the empiricist philosophy, criticizing the associationist perspective on human cognition for its reliance on experience where under the premiss that thought is inherently compositional, insists that only through distancing from experience and through formalization and idealization the productivity of human thought must be explained.

A counterargument to the LoT hypothesis is presented by examining language use where the relation between language and thought is understood under the assumption of language as communication, i.e. a vehicle for publicizing one's thoughts by using linguistic forms , most often in speech, allowing mental life to be demonstrated and evaluated.

From a different angle experience demonstrates that complex reasoning is very much possible without the mediation of complex language. For example, David Gil (2009) argues that Riau Indonesian, a colloquial variety of Malay/Indonesian, spoken by many millions of people as their first language , has a very basic grammar. It is described as 1. morphologically isolating. 2. syntactically monocategorial, 3. semantically associational. There is no reason to qualify the society of Riau Indonesian speakers as cognitively primitive, as by all measures this is a modern human society. Riau is only one of multiple examples of cognitive complexity dissociated from linguistic complexity.

An evolutionary perspective confirms the lack of direct association between complexity in human reasoning and the architecture of language, e.g. Comrie and Kuteva (2005) have argued that concepts usually encoded in grammatical forms in many languages today almost

always can alternatively be expressed in lexical words, and historical linguistics traces a general tendency of transition from concept lexicalization in earlier stages, to subsequent grammaticalization (Heine and Kuteva 2007). So, complexity of thought does not necessarily imply complexity of the language system. This is in unison with the general pattern noticed by sociolinguists that large complex societies, known for conceptual sophistication prefer languages with simpler grammars, while small, less conceptually sophisticated populations usually use language systems with high degree of irregularity, i.e. grammatical complexity. Similarly Bichakjian (1999) argues that languages have not become more complex in the sense of complexity of grammar, but more efficient in packaging in grammatical forms the increasing complexity of their conceptual universe.
In short, there is no clear pattern of correspondence between conceptual sophistication and grammatical complexity.

From a purely theoretical perspective within cognitive science the postulation of LoT, i.e. to ascribe language-like properties to thought is justified as a theoretical necessity to connect reasoning with its instantiation in physical matter, in this case the neurons of the brain, through the mediation of a computational mechanism. Or as per Crane(1990) the postulation of thought as syntactic language serves purely theoretical function.

2.1. Is language really the " mirror of the mind'?

To remind, the generative/biolinguistic perspective argues for a perfect structural alignment between propositions and sentences, between LoT and UG.

" That thoughts and sentences match up so nicely is part of why you can sometimes say what you think and vice versa."(Fodor and Pylyshyn, 2015, p.89, referenced in Reboul 2015).

That said, the stipulation of exhaustive and universal alignment of semantic with syntactic categories is questioned. To remind, the generative/biolinguistic paradigm is a formalism and its scientific value is conditioned on empirical verification, demonstrated only via its use in communication. For example a complete and correct interpretation of the thought, intended by the speaker is made explicit in pragmatic interpretation within a context, e.g. place, time, personal characteristics, e.g. age, sex, profession, cultural backgrounds of the communicating partners. Some concepts and their linguistic representation are only interpretable in cultural contexts, e.g. Santa Claus. Moreover, entire class of linguistic forms, e.g pronouns and indexicals, have meanings only interpretable within the specifics of the current conversation. In this sense as per Gleitman, Papafragou (2013)" if language directly expresses our thought, it seems to make a poor job of it" , because "language just cannot, or in practice does not, express all and only what we mean"(ibid.) In addition, all languages have synonyms and metaphors, a universal demonstration of the lack of isomorphism between concepts and linguistic forms. And more often than not human concepts have no equivalent in linguistic forms and can only be conveyed with non-linguistic signalling, e.g. by rolling one's eyeballs in dismay, or by a sarcastic smile, revealing the true meaning. Significantly, in conversation a complete proposition is rarely verbalized in complete sentences, regularly leaving the deciphering of the intended message to conversation implicature. In fact, very often a proposition is not verbalized at all, e.g. a nod often replaces a sentence “I agree”, a pointing gesture replaces a sentence , e.g. “ Here it is”. Thus, “ language fails to render thought unambiguously”(Gleitman Papafragou 2012). In short “ Language is sketchy, thought is rich” (Gleitman, Papafragou 2013) as linguistic symbols do not align with the units of human cognition and sentences are not a direct and unambiguous representation of propositions. In sum, language is a poor representation of thought.

Given this significant difference in perspectives one is to conclude that one's understanding of the relation of language and thought depends on one's definition of language.

2 .1.1. Diversity of mentalese?

To remind, the LoT hypothesis is based on the stipulation of modularity of mind(Fodor 1975). That said, recent studies have suggested that the argument for a functional specification and spatial isolation of the LoT in a modularized mind is overstated. Mandelbaum et all.(2022) argue against the strict modularity of mind hypothesis and for a strong possibility that the computing abilities are shared with other cognitive faculties, e.g. perception, motor control, vision etc., i.e. computing properties are not restricted to rational thought, but are pervasive within the mind. In this sense a universal computational template, or “ common code” for mentalese is implemented differently in different cognitive contexts by different cognitive faculties for action, perception etc. suggesting the potential existence of multiple varieties of mentalese. And different mental faculties may utilize the universal “ common code” to different degrees, e.g. it is argued that perceptual and motor representations lack the syntactic properties, typical for natural language (Mandelbaum et all. 2022).

Similarly, Gardner, a developmental psychologist at Harvard has proposed a hypothesis of multiple intelligences, arguing that the human brain at birth is configured into 8 different computation modules, or 8 types of intelligence, i.e. logical-mathematical, musical, linguistic, bodily-kinetic, spatial, interpersonal, intrapersonal, naturalistic, although an individual is born with strong innate affinities for mastering one types of knowledge over others (Gardner 1983 and elsewhere). The hypothesis is challenged by many theoreticians, although embraced by educators and therapists.

The modularity of mind hypothesis is also challenged by empirical findings which demonstrate that the brain is an integrated unity, where, although some concentration of cognitive functions for specific tasks emerge during development from adequate experience, these are flexible and adaptable and show no unbridgeable boundaries as the brain functions are adaptable to cope with new cognitive tasks. In this context language is identified as a complex activity where thought and its communication by linguistic forms are integrated(Liebermann 2016 among many others).

2 .1.2. There is thought without language

To remind, under the LoT hypothesis human reasoning is viewed as a symbol-processing activity producing thoughts of the highest level of abstraction. In addition thought is usually understood as necessarily involving the use of language, i.e. rational thought is necessarily linguistic. As per Bickerton “without a system of verbal auxiliaries or verbal inflections there is no automatic and unambiguous mode of expressing time reference. ...Thinking of the kind that humans do is at best extremely difficult in the absence of syntax...” (Bickerton 1990, p.162-163). Thus, complex hierarchical thought is not possible without complex grammar. And the stipulation of the biolinguistic perspective of the mind as a computing mechanism and language as artificial system, reinforces the argument for the nature of human thought as linguistic and abstract.

That said, various scholars have challenged this assertion demonstrating that abstract concepts are formed on the basis of concrete examples and memories of images, visual, audible, tactile etc. and are retained in memory alongside such concrete representations. As one example the language system is learned from concrete examples of language use (Tomasello 2005 and elsewhere). Significantly, these units of of abstract+concrete mental representations are the feed into the computational mechanisms and made available for self-evaluation, i.e. metacognition.

In addition, novel concepts in science, technology, medicine etc. are first conceived and entertained without linguistic labels, a mental processing available to pre-linguistic infants , aphasic humans and animals. Similarly Nida-Rumelin (2010) argues that “ a great part of our thinking is not verbalized” (p.59) and introduces the concept of “ thinking in the broadest sense” to mean “ letting your thoughts wander”, or subconscious stream of thoughts, to include imagination, daydreaming and the like, to contrast with a narrow identification of thought with conscious efforts to rationalize, to include deliberate focus on a problem, rational self-reflection. The later does necessitate mentalese and linguistic representation.

Abilities for processing both conscious and subconscious types of information are available universally to all humans. Moreover, both active, i.e. conscious, and passive, i.e. subconscious, information processing are interconnected as part of the normal behaviour of the mind. Rational thought without the use of language is a common occurrence in various ways and circumstances, and implements both holistic and compositional signs. Non-linguistic rational thought is demonstrated in visual and performing arts, e.g. music, dance, paintings, silent films, graphs, morse code, etc. using non-linguistic symbolic systems. For example, composing a piece of music is a rational attempt to express a thought and/ emotion in musical notes, although it does not involve linguistic concepts. Importantly, metacognition, i.e. self-reflection, does not necessitate verbalization, e.g. recounting one's memorable experiences, imagining future experiences in anticipation, devising a plan for the composition of a painting or a piece of music, etc. and the thinking process of artists is not even verbalizable. So is mathematical problem-solving. In addition, spatial orientation and other practical problem-solving based on deductive reasoning does not necessitate knowledge of language. Dissociation of abstract thought and language is demonstrated by language-deprived deaf individuals, who demonstrate normal cognitive processing and normal propositional thought. In this sense it can be argued that rational thought cannot be equated with language(Nida-Rumelin 2010).

Another, less known and currently controversial, argument that some thought processes do not involve symbols, is authored by Hurlburt, Akhter (2008) who point at the phenomenon and coin the concept “unsymbolized thinking as “ explicit, differentiated thought which does not include the experience of words, images or any other symbols”. It is a distinctly mental, and not perceptual or emotional, experience, a thought, i.e. a short-lived reflection on something, which, as reported, occupies a quarter of human daily cognitive activity.

In addition, developmental psychologists differentiate practical vs. conceptual intelligence,

intuitive vs. rational intelligence, holistic vs. analytic thought (Mounoud 1988).
In short, it is more likely that human reasoning involves a wide variety of thought processes, some of which do not necessarily use symbols and /or language.

2 .1.3. The narrative mind: representing reality as succession of events enfolding in time

For most people narrative, e.g. the creation of myths and tales, is inextricably linked to language and cultural traditions and thus, uniquely available to humans as linguistic species. Moreover, narrative is viewed as essential part of the cultural heritage and cultural identity of a community. In this sense the origin of narrative is conditioned on the pre-existence of spoken language, where language is understood as communication tool.

A challenge to this traditional view of Language First Hypothesis is Ferretti Adornetti 's argument(2020) where the capacity for narration is defined as the ability to organize one's understanding of reality as a succession of causally connected events, unfolding in time by connecting participants, their actions, objects, circumstances. In this sense a capacity for narration , which allows for organizing experiences with reality as hierarchically organized along a time axis, is a capacity of the individual mind. It allows one to remember past events in succession, anticipate likely future scenarios, i.e. Time Travel, and even create fictional stories, designed around the hopes and aspirations of the characters, which entails creativity. In short, the narrative mind is logical and imaginative and narration incorporates two cognitive capacities, for recall of past experiences and for imagination. These two capacities share the same location in the brain, the hippocampus in the temporal lobe, and deficiency in one corresponds with deficiency in the other. Importantly, Time Travel is spatially and functionally associated with Theory of Mind (ToM), both essential contributors to narration processing and there is overlap between the two functions in the prefrontal cortex.

Significantly, under the current consensus Time Travel is uniquely human ability, given its purported dependence on consciousness (Tulving 2005). That said, in Ferretti Adornetti (2020) non-human animals and especially apes " to a greater or lesser extent remember past episodes and anticipate future needs" (ibid. p. 284). Apes demonstrate planning abilities and foresight in manufacturing of tools. Apes also demonstrate a form of ToM as they attribute mental states and ascribe false believes to others and factor these in decision-making, suggesting that the elements of the human capacity for narration have long evolutionary history, supporting a continuity argument. And as per Ferretti Adornetti (2020) a capacity for narration precedes the origin of language, a view outlined in a Narrative First Hypothesis, i.e. the human mind became creative before becoming linguistic.

3. LoT in the human brain and mind

3.1. Spatial and functional dissociation of linguistic computations and logical reasoning in the human brain

Given the assumption of a perfect alignment between the structures of thought and language, this would entail a perfect alignment if not overlap of respective cognitive structures and processing mechanisms in the brain.

That said, these theoretical stipulations are disconfirmed by a number of experimental neuroimaging studies summarized by Fedorenko,Varley (2016) convincingly demonstrate dissociation in the brain areas activated in language processing and a number of other uniquely human cognitive functions. For example, the processing of language and the so called executive functions, e.g. logical reasoning and working memory, is performed in different brain locations by different neuronal networks, as logical reasoning engages resources of general intelligence. There is also dissociation of processing mechanisms for language and general intelligence as they occupy different locations, although both are processed in the left hemisphere. As a highly consequential example, agrammatic aphasics "perform well on complex causal reasoning tasks", ex. play chess, despite significant difficulties in comprehension and production of language (Fedorenko,Varley, 2016). So, damage in language processing areas is not consequential for complex non-linguistic reasoning. Moreover, it is also empirically demonstrated that language and theory of mind (ToM) are spatially and functionally dissociated. For example, the brain areas known to host computations crucial for ToM, are located in the right hemisphere, not dominant in linguistic functions and in aphasic patients ToM remains unaffected (Fedorenko,Varley, 2016). Similarly, compartmentalization of cognitive activities and dissociation of reasoning and grammar in the adult brain is demonstrated in earlier studies of an individual with infection- caused profound damage in areas known for grammar -processing in the left hemisphere, while showing no impairment in rational judgements about reality, mental states, i.e. ToM(Varley, Siegal 2000). Similarly, spatial orientation from environmental cues is processed by mechanisms distinct form language representation as linguistically impaired individuals, e.g. unable to locate objects by following verbal instructions, nevertheless succeed in orientation by using landmark and geometric cues. For healthy people linguistic cues are helpful, but not indispensable in spatial orientation tasks, i.e." language representations are not a mandatory component in reorientation reasoning" (Fedorenko, Varley, 2016).

Interestingly, it is demonstrated that neuronal circuits activated during processing of language are different from those engaged in processing of arithmetic and the inferior frontal cortex is activated when processing structure in language but not in mathematics. In addition, despite expectations for spatial and functional association and even overlap between processing of language and music, given the reliance of both on sound and structure, dissociation in brain topography implicated in processing of structure and in music is found as individuals with impaired processing of music retain normal linguistic abilities and vice versa, i.e. " non-overlapping sets of brain regions are sensitive to the presence of structure in language versus music"(Fedorenko, Varley, 2016).

In sum, linguistic computations , i.e. UG are spatially and functionally dissociated from computations in reasoning. In addition, cognitive activities unique to humans are distributed throughout the brain.

3.1.1.The neurobiology of Merge: a digital operation in an analogue brain

The LoT is conceived as a digital system and the operation Merge as a digital operation, foundational to the functions of artificial systems, e.g. a Turing machine, as it manipulates digital symbols automatically, uniformly and with absolute accuracy. And within the biolinguistic paradigm a digital system is argued to be instantiated in brain tissue, attributing properties of artificial systems to biological matter. Moreover, the stipulations of strict modularity of brain topography and functions and resulting insulation of LoT from the rest of the brain and its functions, is extended to Merge.

Nevertheless, from a perspective of neuroscience brain activity is computational but cannot be digital. As per Distefano (2020) the brain's computations must be understood in a broad sense, where generic computations are understood as sensitive to the properties of the physical system, or the hardware, performing the computations, i.e. the brain as a biological organ. As such, the computational activity of the brain is sensitive to the physiological changes in the biological tissue and cannot be uniform, automatic and perfectly accurate. Moreover, in the reality of bio-cognitive processing LoT and Merge must be integrated withing the overall architecture of the brain as a biological entity, e.g. the functioning of LoT is conditioned on the influx of oxygen through blood supply, energy through nutrition, hormonal activity, the overall chemistry and the electrical activity of neurons, etc., which make brain functioning possible and which could be interpreted as noise and interfering with the computation procedures. Then, as per Distefano (2020) the brain is not a digital organ, as " it lacks the property of digital devices", its operations are analogue. In this sense from a biological perspective there is incompatibility between the two arguments, the two theoretical pillars of the generative /biolinguistic approach, **a.** that LoT is digital in nature, and **b**. that it is part of the human body on par with any other biological organ and system. In short, a digital operation system is incompatible with an analogue biological entity.

In sum, there is dissociation between idealizations and reality. Moreover, a digitalized LoT and the digitalized language faculty as formal proposals have no demonstrable basis in the functions of the brain, i.e. mathematical functions, characteristic for man-made machines cannot be implemented in brain tissue. In this sense the goal of the generative /biolinguistic formalisms , as articulated by Chomsky, to shed light into the internal organization and functioning of the brain and the nature of its computations is out of reach.

That said, consistent with the analogue nature of brain functions, an alternative understanding of Merge is proposed (Distefano 2020) implemented as a non-digital function of non-digital, i.e. analogue computation, where an analogue operation is characterized as "approximation procedure", i.e. less than perfect, where certain margin of error in precision is acceptable and anticipated. In this sense Merge is seamlessly incorporated into the input-output computational routines of natural brain computation.

In short, portraying the functioning of the human brain in terms of absolutes, 1s and 0s as the LoT hypothesis prescribes, is incompatible with the reality of brain functions which are much more complex, nuanced and continuous.

3 .2. Encyclopedic vs. linguistic knowledge: the language / thought interface and its representation and processing by the mind

Human conceptualization of the world, i.e. encyclopedic knowledge, is a systematic organization of our direct experience with the world in terms of conceptual systems. These are only partially encoded in linguistic forms as linguistic knowledge is organized very differently in terms of linguistic categories, some of which are highly abstract and as such are insulated from experience. That said, if language is to function as a mediator for sharing thought, there must be points at which the two systems interface, e.g. material objects are usually represented by nouns, actions are usually encoded in verbs, properties in adjectives and adverbs etc. At the same time there is also dissociation of the two systems, made most obvious in the way lexical words encapsulate encyclopedic knowledge, e.g the grammatical category of noun encompasses animate as well inanimate objects, substances e.g. water, as well as abstract concepts, e.g. justice, law, etc. A wide variety of actions and activities are encoded in the category of verb, and may incorporate manner, e.g. *swim, strangle*, speed e.g. *race, sprint*, etc. (Wolff, Malt 2010). Importantly, encyclopedic knowledge is packaged into linguistic units differently in different languages in reflection of the communities' social and cultural idiosyncrasies, e.g. communities differ in partitioning the colour spectrum, body parts, family relations and the same concepts are encoded in different linguistic categories in different languages. As one example the languages spoken by industrialized societies have larger vocabularies in reflection of a more granular conceptualization of reality by sciences and technology. Diversity in mapping of conceptual and linguistic domains include diversity in encoding relational concepts, such as prepositions and numeral classifiers, etc. Relational concepts are encoded by morphology in some, and by syntax in other languages. Moreover, the meanings of linguistic forms are altered by new generations of learners. In short, there is a "relatively loose fit between language and the underlying conceptual system"(Wolf, Malt, 2010 p.6). This challenges the argument for linguistic meanings as a mirror of LoT or mentalese and for language as a mirror of the human mind.

Nevertheless, there is significant overlap in conceptualization and its expression in linguistic meanings, suggesting innate properties of human cognition, in reflection of the innate perceptual and cognitive properties of human minds. These are demonstrated by universals of colour perception, universally reflected by vocabularies in diverse languages, despite cultural and linguistic differences. In addition, some basic conceptualization of space and motion in infants are explicable with the influence of laws of physics on human cognition (ibid.). This close association of conceptual and linguistic domains is demonstrated by neuroimaging studies, where overlap is found in the activation of neural networks, engaged in processing of perceptual stimuli of shape, colour, motions and the encasing and processing of their expressions in linguistic categories.

In sum, the relation of language and thought cannot be defined in absolutes, e.g. yes vs. no, presence vs. absence. It is nuanced as some aspects of human cognition influence not only linguistic meanings but also linguistic structures, e.g. known information is consistently positioned before new information in a sentence. At the same time the language systems are partially insulated from influences from the laws of nature, through perception and cognition, as they are formed by socio-cultural factors and language-internal processes, e.g. language contact, pidginization etc. So, despite the partial overlap human cognition and language diverge and operate independently under different organizing principles.

For these and other reasons the LOT is generally received with scepticism with the notable exception of Chomsky and some prominent generativists / biolinguists committed to the idea of LoT and UG. This is acknowledged by Fodor and Pylyshyn in the introduction of their book Minds without meanings: An essay on the content of concepts(2013) by stating " Most of this book is a defence and elaboration of a galaxy of related theses which, as far as we can tell, no one but us believes".

3 . 3. Cognition and the Saussurean sign, a challenge of the LoT hypothesis

The sign theory of language by Bouchard presents another challenge to the a LoT hypothesis by defining language in terms of Saussurean signs, an arbitrary association of meaning and form, a signifiant and signifie, a percept and concept. Here the arbitrary nature of lexicon items is determined to be the defining property of language. The formation of the Sassurean sign is understood in therms of computational properties of the human brain (Bouchard 2012) with the intent "to deduce the linguistic computational capacities from the computational capacities that arise at a more basic level of our interaction with the world... the seemingly complex computational properties of language derive from properties of the sign." (Bouchard, ibid. p. xiii). In the context of the sign theory linguistic entities of all sorts, from words to phrases to sentences, are understood as signs. Words are termed as " unit signs " (U-signs) and grammatical markers and syntactic rules as " combinatorial signs" (C-signs). The syntactic properties of language are understood in terms of relations between signs. In this way both lexicon and grammar are framed under common principles. Linguistic signs are also symbols, a specific kind of signs, which means that the relation between signifiant and signifie is arbitrary, based on a convention, i.e. accidental result from social agreement. Moreover, as per the sign theory of language the combinatorial rules, by which C-signs operate forming phrases and sentences, are conventions as well. And syntactic rules, as signs, are meaningful, e.g. the difference in word order has semantic consequence as in "une femme seule" and " une seule femme" (Bouchard, 2012, p. 94).Significantly, the two sides of the linguistic signs are represented in the mind as two types of information, perceptual (acoustic) and conceptual, each represented as abstractions and the connection between the two, which is crucial for the formation of the linguistic sign. The signifies of the U signs form relations of reference while C-signs form relations of reference and predication. To note, despite being conventions, " the set of possible signifiers for C-sign is extremely restricted because the set of physiological relational vocal percepts is small. So arbitrariness is limited by what are ultimately principles of physical science anticipated by biological systems in general" (Bouchard 2015, p.5). So the organization of the language system is influenced by the properties of the human body, contradicting the formal stipulation of language in terms of artificial systems.

4. Challenges to LoT hypothesis from evolutionary linguistics

To begin with, discrete and hierarchical thought exists independent of communication, e.g in hierarchically organized praxis, both instinctive, e.g. in nest building, and cultural, e.g. in tool use by primates and humans. Moreover ontogenetically and phylogenetically there is communication without thought and thought without communication in the human mind, both the developing and fully developed.

4.1. Reasoning in primates

Both thought and communication exist independently in nature, and predate human speciation. And although some aspects of human reasoning remain unique , rational thought, or at least some form of it, can be attributed to languageless creatures, e.g. baboons, chimps and human infants (Ferrigno et all. 2021and elsewhere). As an example, " both human and baboon follow a remarkably similar categorical pattern of responding to *same* and *different* when matching relations, (which) provides unique support for relational abstraction in nonhuman primates."(Flemming et all.2013, p. 523). Baboons have demonstrated analogical thinking, a highly complex thought process, involving detecting structural similarities and mapping abstract relations of two unrelated fields of knowledge. Moreover, great apes' mind registers patterns of co-occurrence and infers cause and effect, suggesting it is capable of Modus Ponens. In addition it registers presence vs. absence , i.e. Modus Tollens, demonstrating ability for two of the very basic cognitive procedures in logical thinking. Call and Tomasello (2005) find that various non-human primates apply complex cognitive strategies in reasoning about navigation, numerocity, causality, sameness-difference, analogical thinking, counting, make inferences about the future on the basis of current information, are capable of creative problem-solving. See also O' Madagain et all.(2022), Ferrigno et all.(2021). Significantly, apes monitor one's own thinking, suggesting a primitive form of metacognition.

Interestingly, although human propositional thought is represented both by images and linguistic symbols, asymmetric representation of the two modes of thought are demonstrated by fMRI experiments, where people apply imagery when thinking about recent experiences and linguistic symbols when reflecting about distant memories. This suggests that non-linguistic thought is preferred when thinking of here-now. Moreover, visual thinking is more basic and primary both evolutionary and ontogenetic timeframes, suggesting innate predispositions, while inner speech using linguistic symbols is a product of learning and conscious control (Amit et all. 2017). So, different species solve cognitive problems by evolving different reasoning mechanisms, which share properties of human reasoning to different degrees, although the topic is largely under-explored.

4 .2. LoT and evolvability

From evolutionary perspective an important tenet of the LoT hypothesis is the assertion that LoT and UG, once emerged, have remained unchanged, i.e. unresponsive to evolutionary forces. Nevertheless, it is natural to envision a role for evolutionary processes and cognitive continuity with primates and perhaps convergent evolution in distantly related species. And given the above referenced empirical findings, demonstrating cognitive continuity, the discontinuity argument for sudden emergence of LoT seems to lack empirical support. Additional support for the continuity argument is provided by Lieberman Ph.(2014) who finds that in response to universal principles of evolution " Existing structures and systems are modified sometimes elegantly sometimes weirdly to carry out new tasks."(p.2). The human brain is not an exception and responds to evolutionary forces by evolving cognitive flexibility in adaptation to changing environments via slight modifications of ancestral brain architecture. For example " We have adapted neural circuits that don't differ to any great extent from those found in monkeys and apes to perform feats such as talking, dancing, and changing the way we act". And we " have identified some of the genes and processes that tweaked these circuits to yield the cognitive flexibility that is the key to human innovation.." (ibid.p. 3). Significantly, the hypothesis for modularity of mind and the characterization of the brain as a digital computer is challenged as Lieberman finds that "operations of a digital computer have no bearing on how real brains work."(ibid. p.3)

To note, the argument for cognitive discontinuity incorporates an argument for cognitive universality, i.e. lack of variation. That said, given the inherent variation in all aspects of the biological organism as a crucial principle in evolution, the argument for universality of reasoning as a biological property of cognition and its hypothesized immunity from evolutionary change, is implausible.

From a different angle, the argument for the beginnings of humanity with a module for logical thought describes the human as introvert and stipulates that the human reasoning is focused on private thoughts, ignoring the environment, physical and social, and in detachment from perceptual and emotional experiences resulting from interactions with nature and human company. In disagreement Tomasello(2014) argues that "human thinking is fundamentally cooperative"(Tomasello, 2014, Preface) i.e. human reasoning and problem-solving is rooted in shared intentions and goals and the uniqueness of human cognition is rooted in human cooperation, demonstrated by human culture, symbols, language and social institutions, as based on communal agreement on common practices, thought and communication patterns. Thus, the uniqueness of human mind is in its ability to make decisions by incorporating other people's viewpoints, i.e. reason collaboratively.

In this sense the evolution of human mind is understood as transformation from the solitary reflection of the ape mind towards " shared intentionality" by coordinated mental states, behaviour and communication. And cognitive propensity to cooperate by participation in joint activities is highly valued in hunting and foraging, assuring its heritability. In this sense cooperation in decision making is the cornerstone of human speciation.

The human collaborative thinking is explicable as a prolonged evolutionary process on the path from the individualistic mind of the great ape and its focus on individual goals , through an intermediate stage of joint attention and dyadic interaction of trained apes and pre-linguistic infants, culminating in the human collaboration of minds and collective decision-making. Tomasello acknowledges that great apes entertain a form of logical thought and even a form of metacognition. And starting form this turning point in cognitive evolution, the modern human mind evolved as the collaborative mind, contributing individual viewpoints, which delivers the unprecedented creativity in culture, language and civilization. The collaborative mind, identified by a membership in a group is reflected in linguistic forms, e.g. the pronoun "we". And although the majority of primate species today are highly social, e.g. they live in groups with internal hierarchical organization in which each member knows its place and participates as a member in the survival, subsistence, defence of the group, humans demonstrate the highest level of sociality among primates, e.g. friendly behaviour is directed not only to family members but to strangers, suggesting that enhanced social skills have been an evolutionary target during primate evolution.

To remind, the argument for a cognitive discontinuity incorporates a hypothesis for a hopeful monster, a highly intelligent loner and introvert with high reproductive potential and his assumed crucial impact on human speciation. That said, such hypotheses are hardly realistic as today geniuses are not known for superb social skills, and an individual with poor social skills would not have been a desirable mate, quite the opposite. Moreover, high IQ does not seam to be a matter of genetic inheritance since high intelligence is not known to run in families.

From a different angle, given the enormous amount of concepts the human mind is capable of retaining and processing in modern advanced civilizations, which, as per the stipulations of the LOT hypothesis is only a fraction of what is innately available as a cognitive potential, this would demand a huge number of genes and much larger genetic code devoted to brain

development. At the same time it is well know that we share 99% of the genetic code with our closest primate species.
In sum, the argument against evolutionary explanation of human rationality lacks empirical support.

5. The argument for the origin of language as Language of Thought

From evolutionary perspective the appearance of grammar as a cognitive property is conditioned on a pre-existing LoT, which imposes its propositional structure upon language, i.e. the propositional structure of thought determines the structure of the sentence. In this sense the computational model of thought is inherited by the computational model of language, supporting the argument for the computational nature of the human mind and emergence of capacity for logical thought and language. Consistent with this position is the biolinguistic perspective that language is primarily an instrument for reflection, identified to be the original function of language, while externalization and use in communication is an evolutionary afterthought (Chomsky, most recently 2016 and elsewhere ; Reboul, 2015).

If this is the case and if one assumes an evolutionary perspective within the frame of evolutionary linguistics, given the hypothesis for perfect alignment of LoT and UG, one would expect a common evolutionary path for these two capacities. That said, as discussed in previous segments, language and thought are neurologically functionally, developmentally dissociated properties of the human mind. In addition, common human experience shows that attempts to verbalize one's thoughts is often a struggle, further confirming the dissociation of the two. In addition, studies in historical linguistics and typology show that there is nothing particularly indispensable about the use of grammatical forms as hierarchically organized conceptual structures can and are, in many languages and in the same language at different time periods , externalized in alternative ways, suggesting an integration of thought and language is best explicable as a cultural phenomenon(Comrie, Kuteva 2005; Heine, Kuteva 2007). But perhaps most significant from evolutionary perspective is the availability of some forms of complex reasoning in apes, suggesting phylogenetic divergence of these two properties. Another challenge to this model is presented by Ferretti (2023) who envisions a multistage transition from silent reflection to communicating by language, featuring an intermediate phase of storytelling by pantomime.

5.1.Narrating thought by pantomime, a gradualist hypothesis for the origin of language

Language is usually identified either as computation, with a focus on structure, or as communication with a focus on meaning. In both these otherwise very different perspectives language is defined in terms of symbol processing, e.g. computation is by definition symbol manipulation and defining language in terms of linguistic symbols is a cornerstone of linguistics going back to Saussure. In this sense the uniqueness of the human mind as LoT and language as UG is squarely placed in the ability for symbolization, identified as the turning point in primate evolution, where motivated signs, i.e. icons and indices in pre-human communication are said to have been replaced in a sudden transition to symbolization and computing abilities, supporting the discontinuity argument.

A challenge to this view is outlined by a continuity argument voiced by various scholars, most recently by Ferretti (2023) who argues that communication by pantomime as an intermediate stage in the transition from private thought and their purposeful communication by linguistic symbols. Pantomime is a theatrical performance using body movements but not language, where information is conveyed by using motivated signs, i.e. icons and indices. This informs the argument for gestural origins of language (Corballis 2003 and elsewhere) given the inherently iconic and indexical nature of gestural signs, consistent with the discovery of mirror neurons by Rizzolatti and Arbib (1998) making possible the transparent articulation of meanings in action frames, i.e. demonstrating the alignment of thought and communication. Ferretii (ibid.) argues that the ability to communicate narrative thought by pantomime is first demonstrated by Homo Ergaster, dated at 1.7-1.4 my, although empirical findings by Genty, Zuberbuhler (2015) identify iconic gestures in bonobos, suggesting much deeper evolutionary roots.

5 .2. Evolvability of Merge, a gradualist perspective

To remind, in generative/ biolinguistic context LoT and its association with UG signalling that semantics is perfectly aligned with syntax as two parallel computational mechanisms. In both Merge has a central role in the computational procedures producing the rational human mind in service of the individual's silent reflection. The externalization of thought by communication is said to rely on a third combinatorial system, the lexicon, by which the hierarchical system of thought is converted into linear order with the material representation in speech, manual sign and writing. In this sense syntax and semantics are considered innate, invariant and universal, while the lexicon is learned and thus, reflective of socio-cultural idiosyncrasies and subjected to cultural evolution. Thus, lexicon and grammar, although interconnected, are understood here as having two very different roles and are products to two very different aspects of the human mind, the former relying on learning capacities, the later on biological endowment. And from evolutionary perspective only the lexicon could be traced to some pre-human bio-cognitive preconditions, while the generative component and Merge, given its presumed accidental emergence, is by definition a unique human property.

A theoretical challenge to this major tenet of the generative/ biolinguistic approach is the proposal by Fujita and Fujita (2022), who envision evolutionary antecedents for linguistic Merge in praxis where motor routines are viewed as hierarchically organized by a Merge-like operation, or proto-Merge, demonstrated in object manipulation by various non-human species. As an example the nut-cracking of chimpanzees is described as hierarchically organized from gestural primitives and labeled as “ action grammar”, first proposed by Greenfield (1991). In this context the evolutionary roots of Merge are found in motor control, argued to have undergone a further evolutionary development in service of communication in protolanguage, by extending its applicability from material objects to ideas and linguistic symbols, an evolutionary step, arguably accomplished by exaptation. To note, dogs and other domesticated species have demonstrated Merge function as they learn very fast to connect a word command with a behaviour, in favour of the continuity argument. In this context Merge is hypothesized as a property of general cognition and termed as “ domain general generative engine” (ibid. p. 408), not unique to humans and to language and utilized by various human behaviours beyond language, e.g. music, mathematics, etc. As for its application in language both lexical words and some meaning-relevant grammatical categories are produced by the same computational procedure of structure generation, e.g. the formation of lexical words and sentences is accomplished by a common generative mechanism centred on Merge (Fujita 2017). Significantly, while lexical words and grammatical categories with referential meanings, e.g. tense, aspect, gender, etc. receive evolutionary explanation tied to their role in meaning representation, the explanation for the formation of abstract grammatical categories, e.g. complementizer, agreement markers and others with no referential contribution are envisioned as products of grammaticalization with the function of maintaining the structural integrity of the language system.

In sum, both the purely cognitive and the communicative functions of language are attributed to a single cognitive mechanism which generates hierarchical structures. And although this novel approach is just a formal proposal, it is preferable to alternatives for the following reasons: **a.** eliminates the characterization of language evolution as a mystery, **b.** offers an evolutionary explanation of language as continuity from pre-human communication and cognition, consistent with the fundamental Darwinian principle, i.e. evolution prefers continuity and domain-general mechanisms which lend themselves to modification.

6. Did human cognition and physiology evolve independently?

To remind, as per the biolinguistic argument the crucial stage in human evolution is the appearance of advanced cognitive abilities by which concepts are stored and processed by LoT and UG, both computational modules, to which the human superb abilities for silent reflection and logical decision-making are attributed. In this context the fact that today language is used primarily as a system for communicating thoughts and experiences is understood as an evolutionary process by which the language faculty, i.e. LoT +UG, was at some later stage in human evolution coopted for communication via its coordination with human physiology for the material articulation of human conceptual structures. In this way an evolutionary adaptation of the human language faculty to the inherently social nature of humans resulted in the evolution of the human capacity for speech, i.e. a capacity to perceive and produce discrete units of sound. In this way the communication of human internal life became shared with speed and precision, another uniquely human trait, underscoring human exceptionality. In sum, the biolinguistic perspective stipulates that at the onset of human speciation human cognition and human physiology have had independent evolutionary trajectories, before converging at a later stage.

The defining feature of human speciation is our cognitive power, afforded by a powerful brain, its size and functions, suggesting that human evolution is primarily an evolution of cognition, which has produced cognitive discontinuity. At the same time, human bodies are not very different from ape's and even lower species genetically, metabolically, physiologically. In short, continuity with the rest of life forms is much more pronounced in human biology, while human cognition displays clear dissimilarities. The most significant physiological difference in humans is in speech capacities. Thus, there is obvious disparity in human cognitive and physiological abilities. To be sure, many questions about human evolution still remain unanswered but this much is uncontroversial and well accepted.

The realization of human physiological limitations, disproportionate to our cognitive abilities, have prompted the domestication of some animal species, e.g. horses, dogs, onto which physiological demands are offloaded e.g. in hunting, transportation, etc. Human civilization deals with this unbalance by introducing technological innovations.

The disparity in cognition and physiology is pronounced in linguistic communication given that the stream of thought is much faster than the human physiology can convey and one can comprehend many more concepts than one can verbalize, i.e. linguistic comprehension is more extensive than linguistic production. Moreover, given that primates have hierarchical thoughts, but not speech, one would conclude that human cognition and physiology have evolved independently.

To remind, the LoT hypothesis is based on the formalization of mind as modularized (Fodor 1983). Segregationist accounts, traditionally focussed on Broca's area in the frontal cortex, assumed to be the most significant evolutionary innovation responsible for the human unprecedented cognitive powers and the location of UG.

That said, recent studies have revealed that this picture is two simplistic. Broca's region has a broad range of cognitive functions which prompts the term " Broca's complex" (Hagoort 2009). It integrates various types of information retrieved from memory and provides internal organization in music, language, praxis, etc.(Sherwood et all, 2008). This is achieved by continuously integrating new information as it is made available. Moreover, Broca's region has similar functions of integration of perception and motor functions essential in observation, imitation, planning, in macaques and humans (Sherwood et all. 2008) Consequently, deficits and/or damages affecting this part of the brain would impair a number of functions. Moreover, the " mirror neurons" providing link between cognition and communication, are located in Broca's. Thus, Broca's region (Brodmann's areas 44 and 45) is found to have heterogeneous composition and functions. In addition, Lieberman (2016) finds that the basal ganglia, a cluster of subcortical brain structures which incorporate Broca's region, play a critical role in both motor control and cognition, have a fundamental role in both cognitive and motor tasks and brain damage in this area disrupts both activities, e.g. in patients with Parkinson disease. Moreover, basal ganglia, " Play a role in associative learning and in planning and executing motor acts including speech in humans and songs in songbirds" (Lieberman ibid. p.138) pointing at evolutionary continuity of this part of the brain architecture.

And from a slightly different angle, from a purely theoretical perspective, as per the minimalist program it is argued that the computational machinery of language is only turned on when it has access to the lexicon of spoken words, morphemes, etc., on which the operation Merge, i.e. external Merge, acts and combines as per the minimalist architecture of LF. Moreover, every additional computing operation which builds additional complexity to the structure , including internal Merge, is only possible with constant interaction with the lexicon which implies phonological component, i.e. externalization for a communicative use. Thus, every product of Merge at every successive step must be externalized by potentially engaging physiology in speech. So, the attachment of speech i.e. the lexicon, conceived as apparent evolutionary afterthought, is really the essential starting point without which the computation process would not produce anything if not for its complete interpenetration with speech at every step. I have not found any explanation for this obvious theoretical inconsistency.

6 .2. Human physiology, a vehicle for advertising thought in linguistic forms

Social animals and especially humans have evolved to instinctively share their private thoughts, emotions and attitudes with conspecifics by using the biological body as a vehicle. And for humans, as highly social species, sharing one's private thoughts and experiences is practically irresistible for which the human body is a powerful instrument. In this sense a legitimate topic of inquiry is the initial modality of language at its inception. Some have argued that a form of open-ended lexicon of symbols materialized by the gestural modality may have pre-dated speech, i.e. the gesture-first hypothesis(Corballis 2003 and elsewhere). Alternatively it is argued that complex, although not necessarily discrete vocalizations, were initially used for communication e.g. musical protolanguage(Mithen, 2007). Both arguments have their supporters and opponents. The gesture-first argument is challenged by the fact that ape communication is multimodal and apes have some articulatory flexibility in vocalizations (Wich et all. 2009). The argument for musical abilities in apes is equally difficult to sustain given the total absence of vocal imitation and vocal cooperation, a challenge for a continuity argument as the human capacity for music demands adaptations different from language, e.g. sensitivity to rhythm, conscious rehearsal, coordinated vocal control, not demonstrated by apes, suggesting that musical capacities must have evolved more recently and perhaps are unique to humans.

And since ape communication today is multimodal, one is prompted to argue , after (Pika, Liebal et all. 2005, Slocombe, Waller and Liebal 2011)for multimodal communication to be the initial vehicle for advertising thought and a precursor for the evolution of spoken language. That said, multimodal communication imposes high energy demands on physiology, which makes it inefficient vehicle for making the fast stream of human thought explicit and exposes the disparity between cognition and communication. This situation eventually leads to the formation of a bottleneck, favouring vocalizations, a much less energy-demanding physiological activity. And given the communicative advantages of vocalizations against gestures, e.g. by freeing hands and facilitating communication at a distance, in addition to being energy-efficient, which explains why speech is a universal default modality, while sign languages are used only when speech organs are damaged. Thus, at some point of human evolution human physiology has undergone evolution as adaptation to cognition.

Such changes would have been initiated at behavioural level from an internal reorganization of the multimodal signal by a process of self-organization(de Boer 2017 and elsewhere) by increasing the articulation precision of vocalizations by learning , resulting in the formation of discrete units within the vocal signal, marking the beginning of speech. Further, as capacities for speech became an evolutionary target, bio-cognitive adaptations would have followed at least partially internalizing previously learned vocal behaviour, resulting in permanent changes in the human body and mind as adaptation to speech.

An evolutionary process known as the Baldwin effect, proposed by the Extended Evolutionary Synthesis (EES) (Jablonka , Lamb 2005) by which certain key aspects of learned behaviour could be internalized by minimizing the needs for learning (Depew 2003), is a good candidate for theoretical framing in an evolutionary explanation of the human bio-cognitive attributes specifically tailored to speech. But, importantly, the Baldwin effect stipulates the historical priority of learned behaviour, which assumes extensive learning abilities, suggesting that cognitive adaptation for extended learning would predate physiological adaptation. In this sense one would envision increased number of meanings being expressed by ancestral vocal capacities, which initially would have been accommodated by behavioural adjustments, would have at some point overwhelmed the physiology of vocalization, creating demands for adaptation. A subsequent adjustment of vocal capacities for speech production would have followed given the articulatory advantages of speech over the multimodal channel, resulting in a meaningful reduction, although not complete, displacement of multimodal signs by speech, as illustrated by spoken dialogues. In spoken dialogues, the most clear demonstration of natural language, linguistic utterances are accompanied by non linguistic multimodal body signals, e.g. manual gesticulations, body posture, vocalizations, facial expressions etc. which clarify and complement the intended message.

The timing of these processes and their association by use in language is difficult to determine given that the ability to produce complex vocalizations, presumably some form of speech, was traced as far back as 400,000ya to the common ancestor of Sapiens and Neanderthals and even earlier as " with a monkey-like vocal tract it is probably already possible to produce a range of articulations that is sufficient to be usable in language"(de Boer, 2017, p. 161), suggesting that some form of communication involving speech, perhaps a primitive form of language, may have predated human speciation.

7. The relation of language and thought has evolved

Language and logical reasoning are two of the most unique human traits. And while today for the literate human it is difficult to separate one from the other as we are used to identify thought with linguistic symbols, the relation of language and thought has evolved as part of human evolution.

7.1. Tool use and symbol manipulation in primates

For understanding the evolutionary roots of human cognitive abilities studying the cognitive traits of primates as our closest living ancestors is of primary importance. Monitoring primate tool use in the wild and symbol manipulation in captivity discussed by Greenfield (1991) reveal that both activities are processed by a common neuronal substrate allowing for a primitive hierarchical assembly of objects by a rudimentary grammar of action, e.g. in tool use on an object, and in symbol combination in gestural language-like communication by using 2-3 symbol utterances of lexigrams.

7. 2. Thought and language in ontogenesis

While the cognitive performances of an adult are a demonstration of the interaction of innate capacities and enculturation, the cognitive abilities of infants and very young children reveal the input of nature in its pure form, unaltered by the influence of nurture. In fact ontogenetic development is a crucial stage from evolutionary perspective as phylogenetically related species tend to have similar early stages of ontogenetic development, a general tendency revealing the conservative nature of evolution aiming to preserve continuity.

Consistent with this expectation similarities are found in the early stages of cognitive ontogenetic development of primates and human children, where a common pattern in mind development is noticed. As in young primates, as per Mounoud (1988) at birth the human mind is a general purpose learning tool which becomes compartmentalized and modularized under the influence of experience, i.e. cognitive development begins as “a general process which is shared by all areas of knowledge, as opposed to the line of reasoning which postulates that there are specific processes for each domain of knowledge” (ibid. p.27). For example the first perceptual-motor activities of a newborn are processed as holistic, e.g. in object grasping, followed by a process of decomposition into combinable units forming a complex behaviour and their creative combination or modification of an instrument, by which a complex behaviour is performed. Parallel processes are detected in language learning, e.g. a holistic representation of stream of speech is followed by partition into phonological segments and a holistic stage of single-word for a sentence is followed by use of words as combinable units forming sentences. From this one is to infer that the development both human thought and language begin from the same amorphous stage and follow the same trajectory of segmentation and specialization while absorbing experience. So, development is a process of transformation of an amorphous mind into a modular mind. In a similar vain Greenfield (1991) finds a parallel pattern of processing in object manipulation and language learning in early stages of development where initial stage of minimal hierarchical organization in both activities, i.e. “ children pass through parallel and quite synchronous stages of hierarchical complexity in forming spoken words and combining objects”(ibid. p.538). Such patterns suggest that at the earliest stages of child development there is “a single neural substrate for the hierarchical organization of language and manual object manipulation” (ibid.p.542) which supports a “ hypothesis for a supramodal hierarchical processor” (ibid. p.534), located in the Broca's area, demonstrated in coordinated movements of hands and mouth. As development progresses dissociation and formation of two functionally differentiated areas controlling these activities is formed by maturity by diverging neuronal associations, demonstrated by divergent patterns in behaviour , e.g. increased complexity of language with no analogue in increased complexity of object manipulation.

From evolutionary perspective, although both primates and human children begin cognitive development from the same starting point of primitive hierarchy in object assembly, demonstrating a common phylogenetic roots, primates remain at this level while children outgrow this stage fast, within 2 years, demonstrating the crucial role of heterochrony, or divergent developmental schedules, in related species.

At the same time it is known that human infants begin language learning equipped with some universal categories, representing dichotomies of animate vs. inanimate, human vs. non-human, singular vs. multiple, close vs. distant , presence vs. absence, instantaneous events vs. processes of long duration, measurable vs. unmeasurable substances, etc.,(Dor Jablonka, 2000, p.39-). These universal frames of categorization come naturally to young learners, suggesting possible innate aspects for the fundamentals of human thought. Moreover, humans are innately predisposed to conceptualize some fundamental ontological distinctions e.g. basic colour distinctions, spatial relations of containment, etc., demonstrated by early human development where young children across cultural boundaries display instinctive awareness of basic ontological categories in uniform way, i.e. early in ontogenesis children demonstrate a “universal pre-linguistic ontology” (Gleitman, Papafragou 2012, p.26)

And because early stages of development reveal innate predispositions, unaltered by experience and socio-cultural biases, one can infer that humans are predisposed to conceptualize some aspects of the environment, essential for human survival, and these universals of human semiosis in universal frames of categorization are identified as a common semantic core consistently reflected in all languages(Dor, Jablonka 2000). Thus, at early stages of ontogenesis some innate predispositions for conceptualizing the world around impose their influence on linguistic concepts and forms, marking the initial stages by which thought is beginning to influence and frame the shape of language. That said, such innate categories and concepts have little resemblance with the postulated LoT given the postulation of its algorithmic nature.

Further, extended experience with language results in greater influence of linguistic categories on categorization of reality, i.e. " long term use of language influences ontology"(Gleitman Papafragou 2013 p.509) by readjusting the conceptual boundaries imposed by linguistic categories, i.e. the conceptual mapping of reality is reshuffled to fit linguistic boundaries determined largely by socio-cultural factors influencing language systems, e.g. in classifying countable objects and substances according to grammatical distinctions is language-particular, some derivative colour distinctions are language-particular. Thus, at early stages of human development universals of human conceptualization guide language learning, while at subsequent stages experience with language expands human conceptual arsenal and imposes conceptual boundaries determined by the local sociolect.

In short, the relationship of thought and language changes during development. In this sense the understanding of the development patterns for thought and language helps understand their relationship from evolutionary perspective.

8. How language shapes thought

An initial stage of narrative by pantomime proposed by Ferretti (2023) envisions dominance of narrative thought over communication. The transition from motivated signs to symbols in communication and specifically the formation of linguistic symbols initiates a new stage which begins to alter the relation of language and thought in favour of language.

8.1. The evolution of the Sausurean sign

At the earliest stages language has originated with iconic signs given the demand of communicative transparency in the absence of established linguistic conventions. As the demand for more information increases at subsequent stages of evolution, the functions of iconic communication are extended, which stretches human physiology to its limits and reveals the inherent limitations of iconic communication. As iconicity becomes an obstacle to the elevated demands for speed and precision due to sloppier articulation in speech and gesturing in manual signs, this triggers an internal restructuring in the communication system by transformation of iconic signs into symbols by eliminating the details, not essential for the accurate material representation and recognition of the concept denoted. As a result the natural connection between signifiant and signifie is completely lost and the icons are replaced by symbols. The dominance of symbols is the final stage in the "gradual transition from the iconic to the arbitrary ground, which does not necessarily entail a sharp gap between the two forms", given that linguistic forms, both lexical and grammatical, retain some level of iconicity(for more see Haiman 1983; Perniss, Vigliocco 2014).

Symbols help in creating discrete concepts and sharpen semantic categories. It has been argued that the use of symbols has had a major role in defining the boundaries of the concept, as " Previously fuzzy properties become sharper after symbolization. **"** (Donald 1991, p. 219 **).** Thus, the formation of symbols alters both meaning and form of linguistic signs, on one hand by streamlining of form by eliminating unnecessary detailing , and on the other enhancing the boundaries of meanings in major contribution to the efficient encoding of the expanding conceptual universe.

The process of transition from motivated signs to symbols is observable in modern times in the formation of sign languages where iconic gestures dominate initial stages of formation in ASL and the Al-SAYIID sign language which later gradually become simplified and lose the natural connection by transitioning into symbols. A similar stage in vocal communication rooted in the need to assure adequate reception of the message by placing limits on the possible interpretations in situations of displaced reference is noticed by Bickerton who argues that the first words would have had to be iconic signals, e.g. imitations of natural sounds (Bickerton, D. 2009). Moreover, the history of writing systems from pictographic images to symbolic representation of various levels of abstraction, e.g. by letters of the alphabet , Chinese characters, etc. suggests that the replacement of motivated by symbolic signs is a general tendency in the evolution of communication. At the same time the fact that language retains some form of connection with the physical reality through the use of motivated signs is a rejection of the strong argument for linguistic determinism over thought and in favour of replacing it with a weaker version, i.e. that language has a role as it influences to an extent the formation of thought, where such influence is most pronounced on thought distanced from perceptual reality, e.g. in imagination and planning.

At the dawn of human speciation the formation and use of symbols was most likely a product of cultural evolution, which likely has triggered significant cognitive adaptation by evolving a developmental capacity for symbolic reference, a complex combination of various types of referential relationships among symbols as members of a symbolic system, between a symbol and its referent, and among objects in reality as perceived by the human mind (Deacon 1997), another significant stage in the evolution of language. Similarly, innate predisposition to form and process Saussurean signs is argued by Bouchard (2012, 2015). In addition, notable biological alteration in the volume and topography of the brain in terms of significant expansion of the frontal lobe as the most significant contribution to the overall enlargement of brain volume of about three-fold, as compared to the ape brain, is interpreted as causally related to symbolization.

In sum, the invention of the Saussurean sign transforms the relation of thought and language and initiates the domination of language over human thought by encasing it in linguistic signs.

8. 2. Grammar is designed to shape human thought

Human concepts are organized into structured combinations and form propositions. In this context Schoenemann (1999, 2005) , a proponent of a usage-based perspective argues that the structured nature of the extralinguistic reality, as reflected by the human mind, determines the hierarchical structure of phrases and sentences in language. Thus, the structure of thought aligns with the structure of language. Nevertheless , in a significant departure from the generative/biolinguistic position here thought is envisioned not in terms of LoT, as it is firmly based on reality and its perception and conceptualization by the mind. Importantly, here phrase structures are groupings based on meaning relations, thus, grammar arises out of meaning. Similar focus on meaning as representation of reality is the perspective of Cognitive linguistics where the distinction between grammar and lexicon, lexical and grammatical elements of language originates in conceptual structure with the distinction between Conceptual Structuring System, which " provides the structure, skeleton…of a given scene " (Evans, Green, 2006. p.192,) thus, a generalized model of a typical scene, a type, and Conceptual Content System, a token, which " provides the rich detail of a particular scene" (ibid. p.192).

To remind, the language system is a way of representing conceptual structure in a very specific way, through the mediation of semantic structure. Although this structural distinction at conceptual level is relatively vague, it provides the foundation for the formation of the semantic system (thematic roles) on which grammar is based as a higher level of abstraction.

"semantic structure represents the conventional means of encoding conceptual structure for expression in language". (Evans, Green, ibid. p.192)

That said, the overlap between conceptual and semantic structure is only partial. Not all thought is communicated and what is communicated is not only thought (but also emotions, cultural forms of etiquette, etc.) :

" while semantic structure must, to some extent at least, reflect conceptual structure, and while semantic structure can be thought of as a subset of conceptual structure, a system of lexical concepts selected for expression in language, the relationship between conceptual structure and semantic structure is…complex and indirect." (Evans, Green, 2006, p.193).

In short, the conceptual system does influence, although does not determine, the shape of language.

From evolutionary perspective introducing complex grammar , i.e. grammatical categories and structures with no semantic value , with purely structural role, e.g. complementizer, and structures, e.g. double negation, repeated marking of tense and gender on each adjective within the NP, embedding of phrases and sentences, etc. is a tool for shaping and organizing human thought with the goal of more successful understanding. The role of grammatical devices, from case marking to pronouns, to topicalization of phrases etc. is to make the speaker's ideas and intentions easily understandable by the listener and avoid ambiguity. Question words and their grammatical configurations are only explicable with a demand for information and pronouns, definite and indefinite, signal the difference between information known vs. unknown. In sum, language has evolved to facilitate adequate understanding of human thought.

And from a different angle as per Schoenemann (2005)increase of semantic complexity in the form of increase in the number of words in the lexicon leads naturally to the emergence of hierarchical structure in grammar, i.e. syntax evolved as a result of expansion in semantic content which leads to restructuring of its encoding and communication. That is, the formation of grammar is a solution to the overload which conceptual complexity creates for language processing in the brain.

But perhaps most importantly for the current topic under discussion, the here mentioned perspectives deviate from the LoT hypothesis and its computational approach to the human mind and its isolation from experience, and on the contrary underscore the influence of perception and experience on human mind and language.

8.3. The dominance of language over thought in writing

Thought shaped by writing is quite different from thought shaped orally. It introduces a structure quite different from that of the dialogue, demonstrating that education has influence over the structure of thought by stimulating the formation of abstractions. Dabrowska (1997) confirms an obvious fact of everyday experience that people with higher education, especially those for whom language is a professional tool, are able to comprehend and produce language of much higher complexity using highly abstract grammatical concepts and forms as compared to manual labourers.

Literacy is also suggested to influence the structure of one's memory, e.g. illiterate individuals' memory storage contains more concrete examples, whereas literate people organize their memories by using abstractions. The influence of literacy and education also alters perception where is language represented by the technology of writing in terms of strings of discrete spatially arranged characters , e.g. letters of the alphabet . For example, the perception of the sound stream as a string of phonemes and the understanding of the phoneme as an abstract concept is influenced by experience with writing (Port, 2007, p. 153). Literacy is found to influence not only cognition and perception but also brain connectivity and processing as "the brains of literate persons are substantially (differently)rewired compared to that of their illiterate siblings"(Levinson, 2012, p. 397). Similarly, Ardila et all.(2010) argue that exposure to literacy and education influence brain activity and result in changes in brain connectivity and potentially brain lateralization. For example fMRI tests show increased activation of the left hemisphere in brains of individuals with post secondary education, compared to illiterate individuals where the right hemisphere is more engaged. And exposure to literacy and education is argued to stimulate executive functions by directing focus on one's mental activity, i.e. metacognition, making it an object of reflection and judgment.

Thus, formal education, which comes with extensive engagement with written texts, is a crucial factor influencing significantly the linguistic behaviour of speakers and alters the relation of language and the mind.

And from a slightly different angle, given the diversity of languages and diversity of writing systems, a potential role for the diversity in sociolects in creating diversity in thinking, a form of the Sapir-Worf, is not to be totally dismissed.

8 .4. Articulation of metacognition in language

Metacognition is a cognitive ability of the highest order of complexity, including an ability to evaluate mental representations, removed from the here-now direct experience, monitor and register inadequacy of knowledge, to actively supervise and deliberately self-correct, which implies a mind with self-consciousness (Heyes, 2012) i.e. it is thinking about thinking. Moreover, humans are also highly social species and social intelligence, or Theory of Mind (ToM) i.e. the ability to correctly detect and anticipate the thoughts, desires, aspirations, of fellow humans, is understood as integral part of high cognitive functions. ToM is found to be instrumental in metacognition, e.g. human metacognition, the ability to exert control over one's decisions, is conditioned not only on knowing one's own mental and emotional states, but also the mental and emotional states of others, given that the individual's mental states and behaviours are intertwined with those of fellow community members. (Papaleontiou-Louca 2008; Metcalfe, Schwartz 2016).

And although a " mentalizing capacity"or Theory of Mind of some primitive form is available to languageless creatures, exploration of mental life as a focus of attention is greatly facilitated by the language system, which provides the natural connection between recursive thinking and communication, exemplified in the use of verbs which require sentential embedding , e.g. *believe, think, suppose, imagine, guess,* etc. (Oesch and Dunbar 2016), linguistic forms most often used in writing, which encourages the use of abstract vocabulary and structural sophistication. Significantly, it is argued that the very concept of self , i.e. the sense of having individual identity, responsibility and authorship of one's behaviour and its implications, is tied to its explicit articulation in language. In addition, language influences one's reflection upon oneself, makes it easier to identify one's mental states, define clearly individual goals, share information and agree on joint future projects. In this sense it has been argued that the properties of language systems influence one's thought process in decision-making , including one's self-reflection, which is why, although a number of scholars have attributed episodic memory to primates and even lower mammals(Tomasello 2023), assessing the mental life of languageless creatures is difficult.

Significantly, the fact that metacognition is encouraged and shaped by linguistic forms is the clearest example of dominance of language over thought.

8 .5. Evolvability of Merge: evolving digital infinity by cultural evolution

The generative/ biolinguistic perspective identifies the language algorithm as a digitalized language machine with bio-cognitive dimensions with Merge operation having a critical role in both LoT and UG. As such Merge maintains the crucial properties of the language machine, i.e. innateness and universality, given the proposed explanation for its existence involving a biological accident, i.e. precluding a possibility for evolutionary continuity.

That said, as argued in previous segments, a form of Merge, i.e. proto-Merge, is a cognitive property available to various non-human minds as they reflect the material, physical and biological reality in which we and the rest of biological forms live, reflected by the animal mind, e.g. facts of life, death, birth, food, starvation, etc. by Merge-like although non-digital mental operations. And given the continuity argument such mental operation is also innately available to human minds inherited by evolutionary processes.

In this sense Knight, Power (2008) argue that a digitalized form of Merge is within the capabilities of human minds, although it is not innate or universal but a product of cultural evolution, where the physical reality is augmented by the artificial reality of human culture and the institutions, constructed by human civilization (Searle 1996), e.g. money, marriage, religion. In this sense cultural inventions are perceived by the human mind as facts. And digitalization is instrumental in the formation of the human-made reality and vital to science and technology, thus, Merge has evolved within human culture, not by biological processes, suggesting that it is learned and subjected to cultural differences, i.e. it cannot be an innate, thus, universal, property of the human cognition. In this context the ability of the human mind to entertain digitalized concepts and their application in language, understood as communicative technology, and especially in writing , is a product of learning and adaptation of the flexible human minds to the digital nature of the artificial reality as product of cultural evolution.

In sum, civilization and writing has altered the relationship of thought and language in favour of language.

Summary and conclusions

The relation of language and thought is a controversial topic and arguments are shaped by scholars' definition of language. The generative/ biolinguistic position incorporated the LoT hypothesis in its computational approach to human mind and language. In this context the algorithms organizing human reasoning and language are aligned and function in perfect unison, i.e. language and thought are indivisible.
But at a closer look has revealed that this relationship to be highly complex as not all thought is linguistic and not all content communicated by language is thought. In this sense although it is undeniable that thought and language are intimately intertwined as the main purpose of language is encoding and sharing thought, the relation cannot be defined as perfect alignment of computing processors but as mutual influence.

From evolutionary perspective thought predates language both phylogenetically and ontogenetically suggesting that these two capacities have followed by separate evolutionary paths. This, at the same time indicates that thought and communication exist in the human mind independently and interact only in specific circumstances in which such interaction is determined by a person's conscious decision. At the same time over the course of human evolution the human body and mind have evolved cognitive and physiological attributes to facilitate the interaction of reasoning and communication. Further with the emergence of civilization and with it the proliferation of literacy this relation has changed significantly. Most notably under the influence of writing and with it the repositioning of human language in close resemblance to artificial systems, the thought of literate humans is reshaped to align with the algorithmic nature of written language.

In conclusion language and thought mutually influence one another, i.e. the relationship is bidirectional. But this relationship has evolved during human evolution. Thought precedes language evolutionarily, but once evolved, language channels and even shapes thought.

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